

Artificial Intelligence

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Overview

- Course Structure
- Artificial Intelligence Syllabus
- Job Ready
- Lecture

Course Structure

Artificial Intelligence (AI) Syllabus

1. Introduction (4 hours)
2. Problem Solving and Search (9 hours)
3. Knowledge Representation and Probabilistic Reasoning (7 hours)
4. Machine Learning Fundamentals (10 hours)
5. Neural Networks and Deep Learning Algorithms (8 hours)
6. AI Applications (5 hours)
7. Emerging Trends (2 hours)

Marks Distribution

Year/Part: III/II

Teaching Schedule				Examination Scheme						Total
L	T	P	Total	Theory			Practical			
				Assessment Marks	Final		Assessment Marks	Final		
					Duration (Hrs)	Marks		Duration (Hrs)	Marks	
3	1	1.5	5	40	3	60	25	0	0	125

Chapters	Hours	Marks distribution*
1,6 and 7	13	16
2	9	12
3	7	10
4	10	12
5	6	10
Total	45	60

Job Ready 🙌

Industry Ready Skills

Programming skills

Python, C++

Data Visualization

Pandas, Matplotlib

Seaborn, Numpy

Machine Learning Framework

Scikit-learn

Deep Learning Framework

Tensorflow, Keras, Pytorch

Data Management and Engineering

SQL, Hadoop, or Apache Spark

Model Deployment & MLOps

Docker, Kubernetes, Git, and cloud platforms (AWS, Azure, or GCP)

Prompt Engineering

Large Language Models (LLMs) and other generative AI

Senior AI Solution Engineer

TalentSathi - Simplifying the Recruitment · Kath...

 Easy Apply

Save



Responsibilities of the Candidate:

- Lead the **design, architecture, and deployment** of advanced AI solutions.
- Develop and optimize **LLM and NLP-based** systems for production environments.
- Work extensively with **agentic frameworks** like crewAI, langgraph, or similar orchestration tools.
- Build and manage **backend systems**, including API development and database integration (graph, vector, SQL).
- Oversee deployment, scaling, and monitoring of AI systems on **AWS, GCP, or Azure**.
- Provide **technical mentorship** to junior and mid-level engineers.
- Collaborate with cross-functional teams to align AI solutions with business needs.
- Explore new AI technologies and tools to continuously improve system efficiency and capability.

Requirements:

- **4+ years of professional experience** in developing and deploying AI-based systems.
- Proven expertise in **NLP, LLMs, and machine learning workflows**.
- Experience with **agentic frameworks** (crewAI, langgraph, etc.).
- Strong background in **Python**, AI frameworks, and **API design**.
- Hands-on experience in **cloud deployment and management** (AWS, GCP, Azure).
- Proficiency in **database technologies** (vector, graph, SQL).
- Excellent leadership, analytical, and communication skills.

Jr AI Engineer

SCAN · United States (Remote)

Apply 

Save



Your Qualifications

- Bachelor's Degree or equivalent experience preferred
- Foundational understanding of LLM concepts, prompt engineering, vector databases, and how Azure AI Services work.
- Basic proficiency in Python (or similar), REST APIs, JSON schemas, and workflow automation tools (Logic Apps, Power Automate, n8n, etc.).
- Familiarity with Azure AI Foundry fundamentals (model catalog, agent service, connections to Azure AI Services) is a strong advantage.
- Strong curiosity, initiative, and ability to absorb feedback in a fast-moving, experimental environment.
- Commitment to Responsible AI practices—understanding transparency, traceability, and nice to have basic bias/fairness awareness.
- Collaborative spirit; able to support multiple initiatives and interface with business, analytics, and engineering teams.

AI Researcher Skills

Domain Knowledge

Healthcare, Building Structure
Monitoring, Material Science,
Computer vision (CV), Natural
Language Processing (NLP) etc

How can we get Domain Knowledge?

Path way masters, and PhD

What Concept we need?

Must of the part is similar to
industry skills

What makes researcher different?

Researchers create new
knowledge.

Industry engineers apply existing
knowledge.

Mathematics & Statistics

Linear Algebra, Multivariable Calculus, Probability, and Optimization (e.g., understanding gradient descent)

Theoretical Foundations:

Deep knowledge of various AI sub-fields, such as Advanced NLP, Reinforcement Learning, or Computer Vision

Specialized Tools

academic tools like LaTeX (for writing papers)

What does matter in research?



What actual Quality work looks like?



Yann LeCun

Chief AI Scientist at Facebook & JT Schwarz Professor at the Courant Institute, [New York University](#).

Verified email at cs.nyu.edu - [Homepage](#)

[AI](#) [machine learning](#) [computer vision](#) [robotics](#) [image compression](#)

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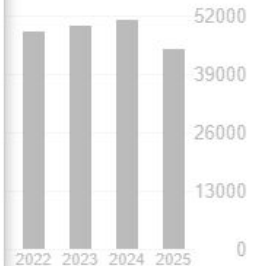
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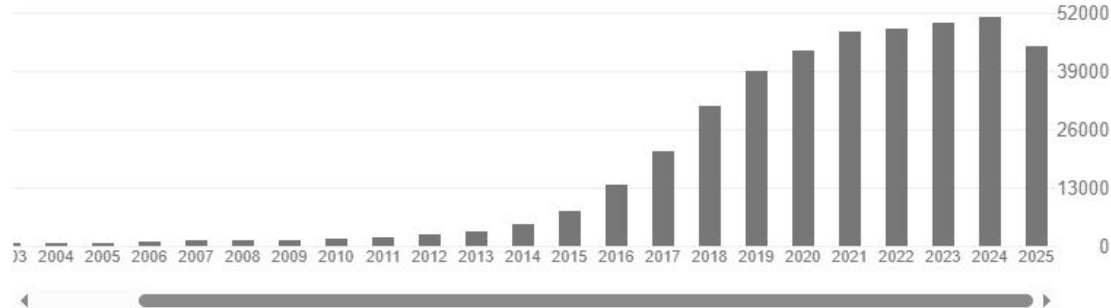
All

Since 2020

437101	285961
164	128
443	352



Citations per year



Journals' Rankings

Q1

21% 6

— based on 28 most cited papers

TITLE

Deep learning

Y LeCun, Y Bengio, G Hinton

285 [Q1](#) [NA](#) [nature 521 \(7553\)](#)

Gradient-based learning

Y LeCun, L Bottou, Y Bengio

47 [Q1](#) [NA](#) [Proceedings of the](#)

Backpropagation applied to handwritten zip code recognition

Y LeCun, B Boser, JS Denker, D Henderson, RE Howard, W Hubbard, ...

30 [Q1](#) [NA](#) [Neural computation 1 \(4\): 541-551](#) [USA](#)

Public access

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0 articles

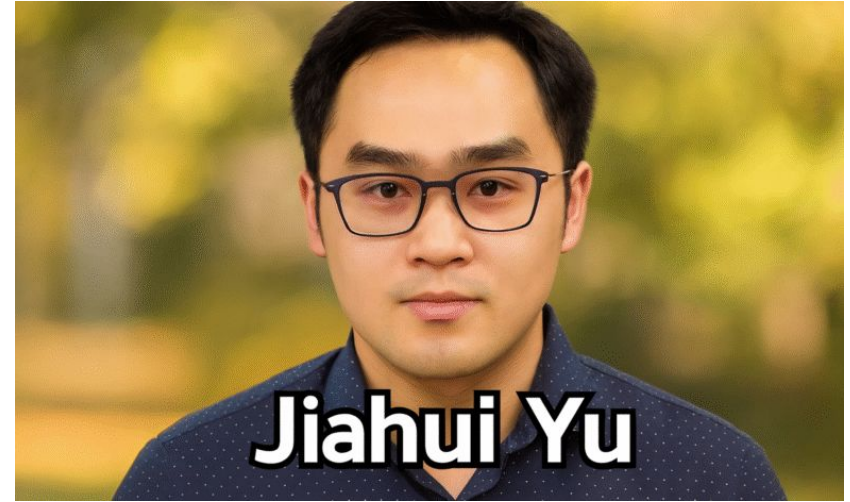
[21 articles](#)

Football Player



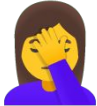
\$80 million

AI Research Scientist



\$100 million

Who is real celebrity?

Exam Ready 

Intelligence

Intelligence is the ability to process information to solve problems and adapt to new situations.

The word **Intelligence** comes from the Latin “**Intelligentsia**”
(from **inter-legere**, meaning “**to choose between**” or “**to understand**”)

Conscious Mind



Its you

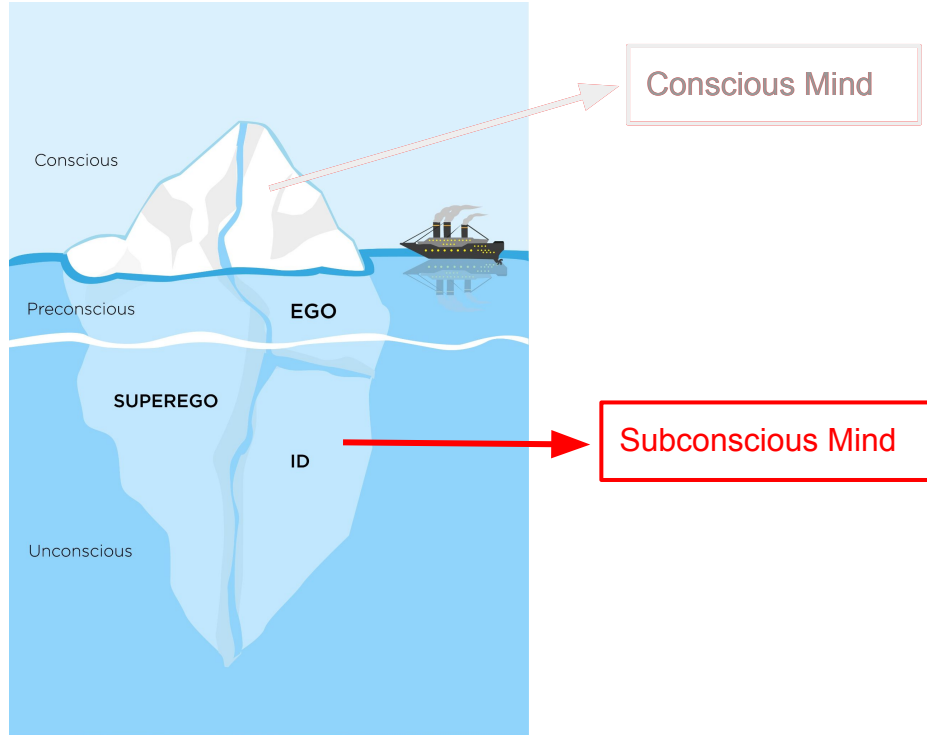
Who is trying to focus on my side and what I'm trying to say?

Or

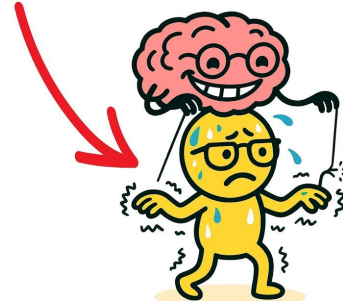
Its someone who is thinking about playing Fifa, watching movie or series etc.

It controls logic, willpower, and critical thinking.

Subconscious Mind



You're Not In Control



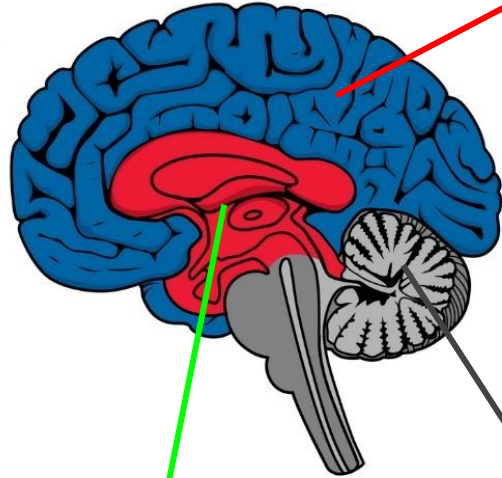
Autopilot Mode

Function: Heartbeat, Breathing, Emotional Reactions, and Habits.

Storage: It stores every memory and experience you've ever had.

"Gut feeling" or "Intuition"

Brain Biology



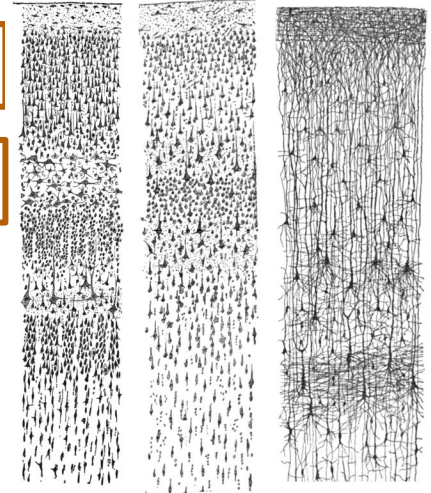
Neocortex:

Logic, Language, Planning, Math.
This is what IQ tests measure.

"Deep Learning" used by
ChatGPT is heavily inspired by the
layered structure of the Neocortex.

Cortical Column:

Microcomputer



Cerebellum:

Movement, Balance, muscle
memory.

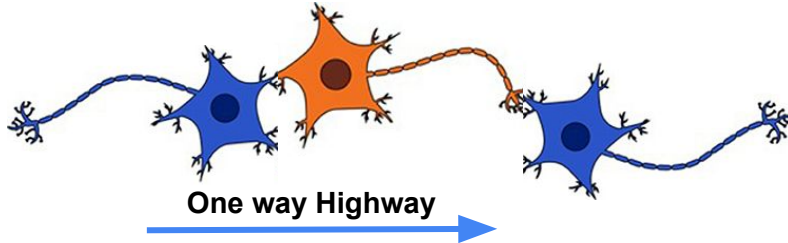
Limbic System:

Emotion, Fear, Motivation.

How Does Our Brain Create Intelligence?

Brain is not a single solid muscle; it is a network. Intelligence arises from connection, not just size.

- The Hardware (Neurons): You have about 86 billion neurons (brain cells).
- The Software (Synapses): Each neuron connects to thousands of others. These connection points are called synapses.
- The Mechanism: When you learn something new, your brain physically rewires itself. It strengthens connections between certain neurons.



Intelligence is the speed and efficiency of these electrical signals.

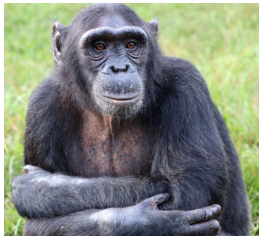
Why Do We Have Intelligence?

Evolution doesn't give us things for fun; it gives us things to help us survive. We have intelligence because we are physically weak.

- We don't have sharp claws, thick fur, or wings.
- We evolved to out-think our environment. We traded muscle for brain power.

Do Other Animals Have Intelligence?

High-Tier Intelligence

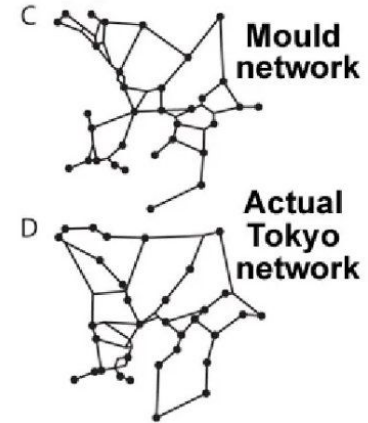
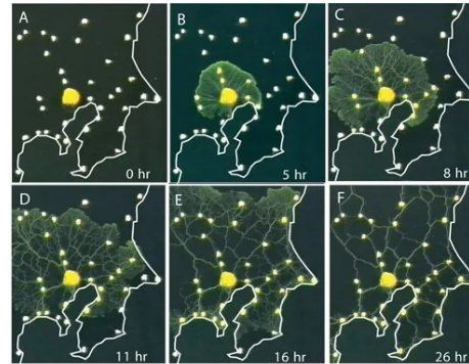


Do We Actually Need a Brain to Have Intelligence?

No. This is the most shocking discovery in modern biology.



Slime Mold



Slime Mold Beats Humans at Perfecting Traffic Networks

So, you do **not** need a **brain** to be **intelligent**.

Formal Definition of Artificial Intelligence

- McCarthy, Minsky, Nathaniel Rochester and Claude E. Shannon coined the term **"Artificial Intelligence"** in a proposal that they wrote for the famous **Dartmouth conference in Summer 1956**.
- “Artificial Intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs” - John McCarthy

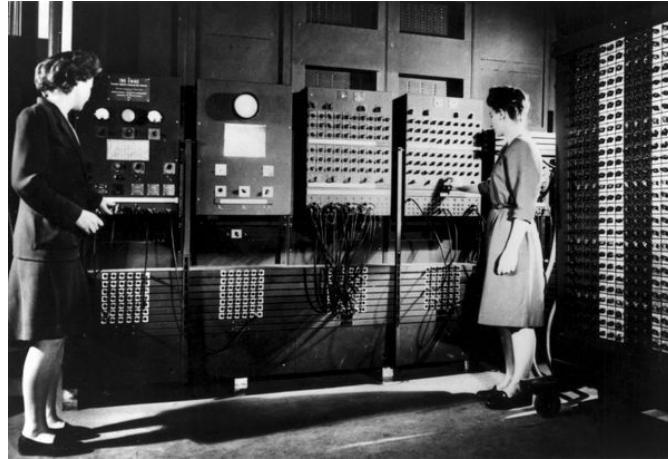
or

AI is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

Brief History of AI

1946: Electronic Numerical Integrator and Computer (ENIAC)

- first programmable,
- electronic,
- general-purpose digital computer
- Turing-complete
- It weighed 30 tons and covered 1,800 sq ft.
- It wasn't "smart."
- physically unplug and replug hundreds of cables.
- It was less "AI" and more "Electric Switchboard."



1950: Turn Over Moment with Single question...

- "Can machines think?" ask by Alan Turing
- Father of Computer Science
- He proposes a test: If a machine can fool you into thinking it's human, it counts as intelligent.



1956: A New Field Was Introduced

Dartmouth Conference

- 2-month, 10-man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire
- In creating the machine to think and improve themselves
- Dartmouth AI Project Proposal; J. McCarthy et al.; Aug. 31, 1955.



Founding Father of AI

70 years later, we are still working on the goals they hoped to finish that summer.

1964 - 1966

1964: Eliza - The first chatbot psychotherapist

One person started chatting with Eliza and Eliza started asking personal question and this person started answering personal question and felt there was a real psychotherapist sitting on the other side

1966: Shakey - First General purpose mobile robot (Stanford)

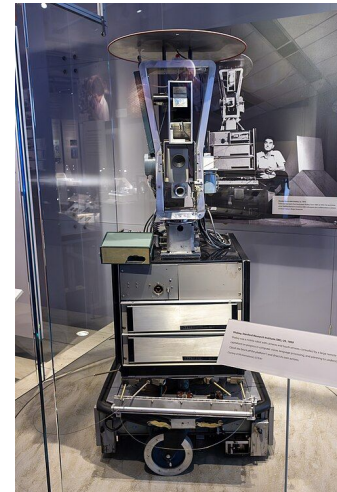
The most important algorithm in AI-
A* search algorithm

```
Welcome to

EEEEEE LL      IIII ZZZZZZ AAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LL      II      ZZ  AAAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LLLLLL IIII ZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:  Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:  They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:  Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:  He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:  It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```



AI Winter

The First AI Winter (1974 – 1980)

- Criticized machine translation (e.g., Russian to English)
- Massive funding cuts by the US government.
- Negative results in Neural Networks (XOR)

The Second AI Winter (1987 – 1993)

- Expert system era- 90's (old style AI course)
- They couldn't learn and required expensive human experts
- Researchers started calling their work "Machine Learning," "Informatics," or "Statistics" just to get funding.

AI Winter

People get stuck in time

People started to cut off the cord with AI, there were topics that were part of AI but created their own sub communities they started saying

- we are not AI we are CV
- we are not AI we are NLP
- we are not AI we are ML

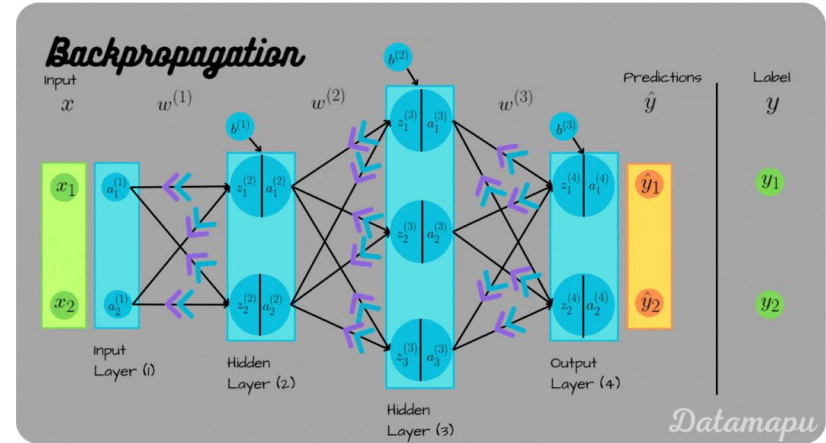
"Hangover" effect of the AI Winters

– [Economist 07] “Artificial Intelligence is associated with systems that have all too often failed to live up to their promises.”

– [Pittsburgh BT06] “Some believe the word 'robotics' actually carries a stigma that hurts a company's chances at funding.”

Backpropagation Algorithm (1986)

- The Breakthrough: **Backpropagation** (popularized by Geoffrey Hinton, David Rumelhart, and Ronald Williams).
- Neuron can update and reduce the error rate.
- Neural networks to have multiple layers (Deep Learning)
- Computers are too slow, and data is too scarce to make it do anything truly useful.



AI in the 90's

1997: Deep Blue (IBM) ends Human Supremacy in Chess

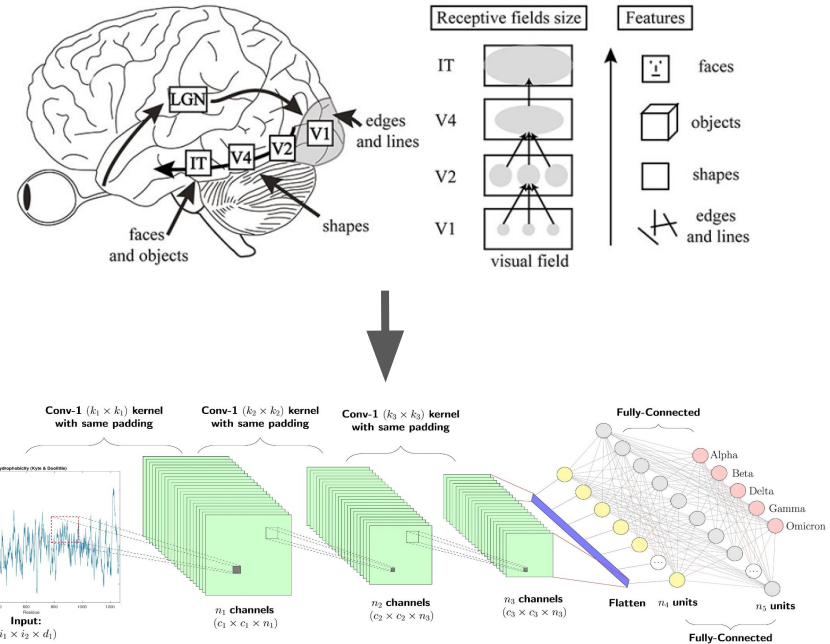
“I could feel human-level intelligence across the room” -Gary Kasparov, World Chess Champion (human)

In a few years, even a single victory in a long series of games would be the triumph of human genius.



1998: LeNet (Now AI Can Start Seeing things)

- LeNet is a series of convolutional neural network architectures
- Research group in AT&T Bell Laboratories
- First applied the backpropagation algorithm to practical applications
- No enough Data available



https://www.youtube.com/watch?v=FwFduRA_L6Q

DARPA Grand Challenge (2005)

- Goal: Drive 142 miles across the Mojave Desert autonomously.
- The Prize: \$1 million.
- The Result: Total Failure.
 - 15 cars started.
 - 0 cars finished.
- Probabilistic Robotics—teaching the cars to "guess" and manage uncertainty.
- Goal: 132 miles across the desert (tunnels, cliff edges, dry lake beds).
- The Winner: Stanley (Stanford University).
 - Time: 6 hours, 53 minutes.
 - Drive faster (up to 40 mph).



<https://youtu.be/LZ3bbHTsOL4?si=T6OSQPMixzy6idJl>

2007 DARPA Urban Challenge

- Whitaker made new car “Tartan” and won the DARPA urban challenge.

https://youtu.be/IULI63ERek0?si=yBOfka6eyvwep_YL

- 12 year anniversary- 2017 got the car back from garage and cleaned up and find out the problem- just one component went wrong and they lost the challenge.

ImageNet (2012)

- The Competition: ImageNet Large Scale Visual Recognition Challenge
- Backpropagation on a massive neural network called AlexNet
- Standard computer chips, but by GPUs—chips designed for video games

**Big Data + Fast GPUs + Neural Networks =
Superhuman Perception**



Network	Error	Layers
AlexNet	16.0%	8
ZFNet	11.2%	8
VGGNet	7.3%	19
GoogLeNet	6.7%	22
MS ResNet	3.6%	152!!

2016: AlphaGo (Google DeepMind)

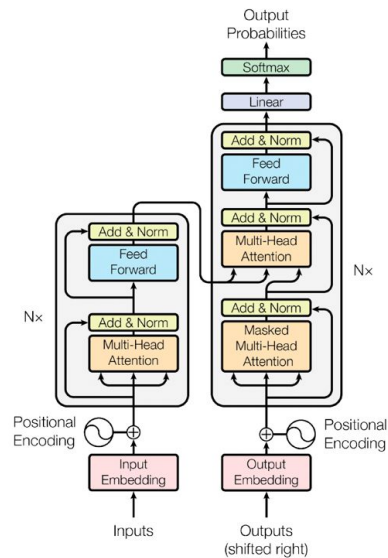
- AlphaGo mastered the ancient game of Go, defeated a Go world champion (Lee Sedol), and inspired a new era of AI systems.
- "Even if I become the number one, there is an entity that cannot be defeated." By Lee Sedol

19 November 2019



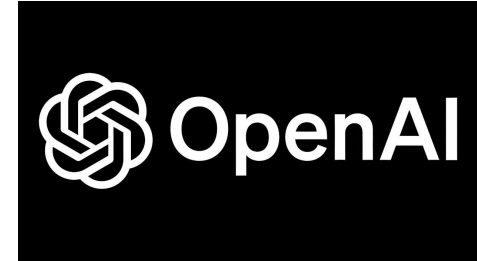
2017: The Language Breakthrough

- Google researchers publish a paper titled "Attention Is All You Need," introducing the **Transformer**
- Understand the relationships between all words simultaneously
- Fill the missing link for human-level language understanding



2022–2023: The Generative AI

- OpenAI released a chatbot that could write poetry, debug code, and explain quantum physics.
- The Transformer Architecture scaled up
- DALL-E 2 appears, creating photorealistic images from simple text prompts.



2025: Spatial Intelligence

- AI to understand the physical, three-dimensional world
- Dr. Fei-Fei Li is the "Godmother of AI"
- World Labs built **Large World Models**
- Fully interactive 3D worlds from a single image

By Yann LeCun, has famously argued that "LLMs are a dead end."

<https://www.worldlabs.ai/blog/marble-world-model>



What is being said about AI?

The New York Times Magazine

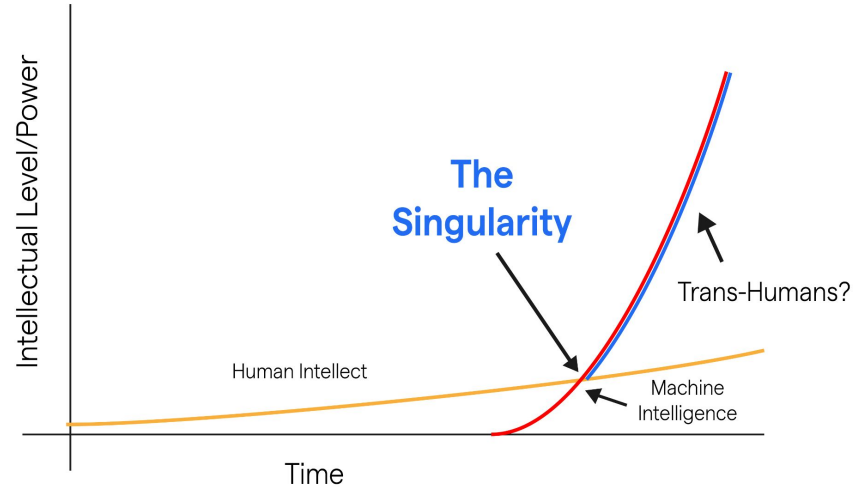
A.I. Might Take Your Job. Here Are 22 New Ones It Could Give You.

In a few key areas, humans will be more essential than ever.

Elon Musk says that in 10 to 20 years, work will be optional and money will be irrelevant thanks to AI and robotics

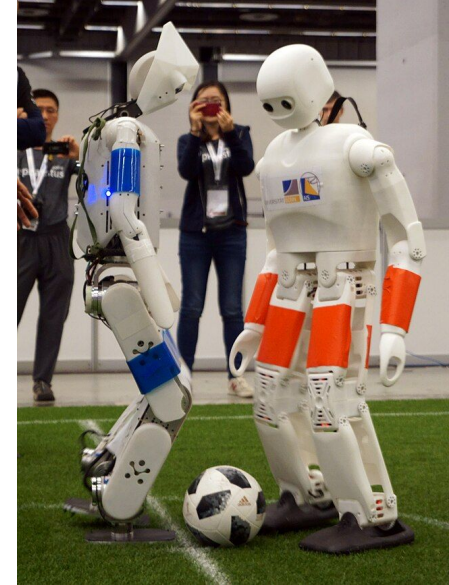
Is This Possible?

**Humanity May Reach
Singularity Within Just 5
Years, Trend Shows**



Or We still lack “I” in AI?

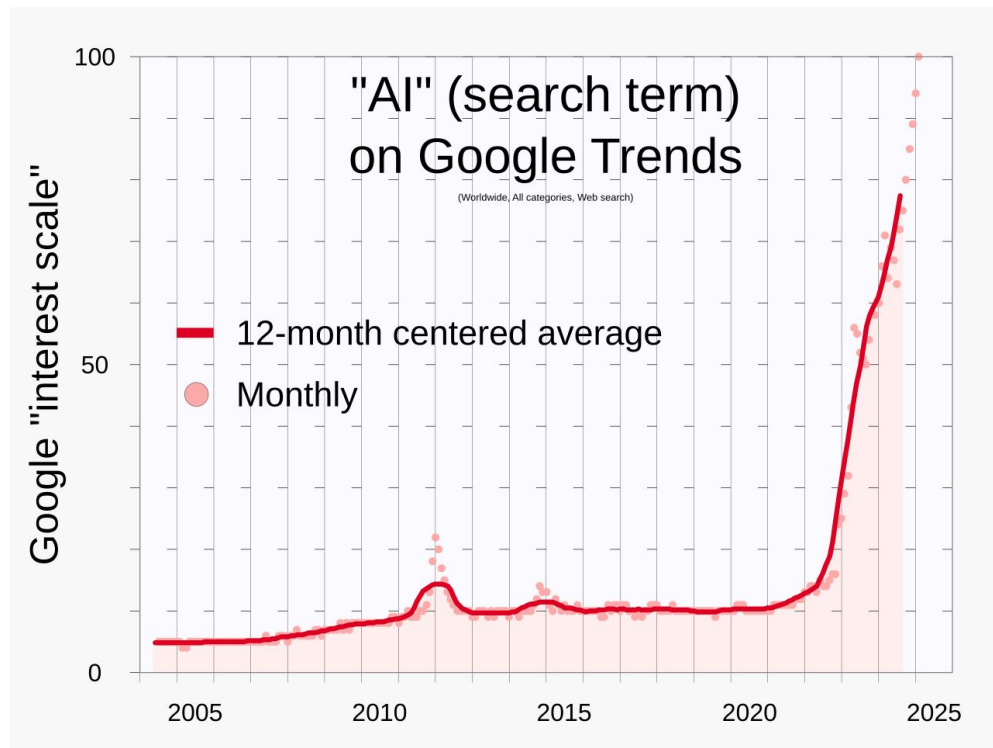
AI Olympics (2035–2050)



<https://youtu.be/pCLD93Uuzac?si=fvNhCf8M8OLQrETI>

<https://www.youtube.com/shorts/Y8VyMQ22YxI>

AI Bubble



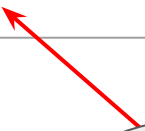
AI

- Historically, researchers have pursued several different versions of AI.

Human Like vs Rational

Thought
Vs
Behavior

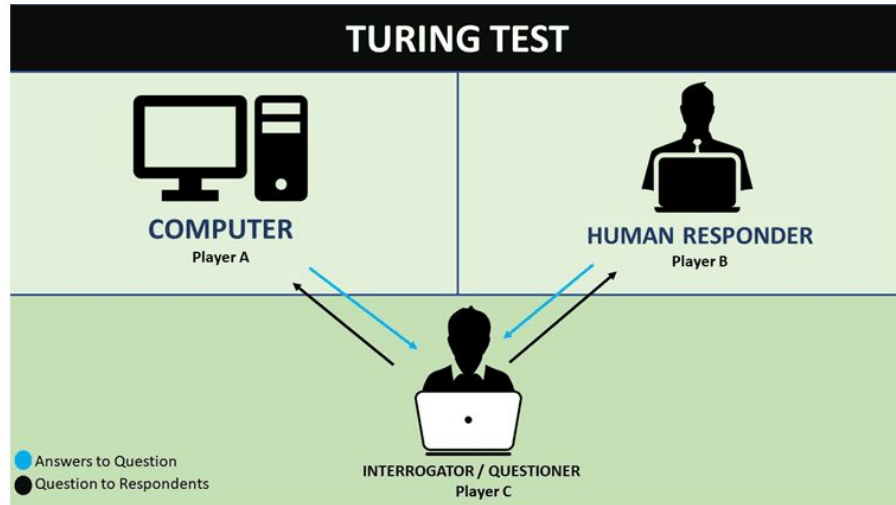
Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally



Winner

Acting Humanly: The Turing test approach

If the human cannot tell whether the responses from the other side of a wall are coming from a human or computer, then the computer is intelligent.



Acting Humanly: The Turing test approach

To pass a Turing test, a computer must have following capabilities:

- **Natural Language Processing:** Must be able to communicate successfully.
- **Knowledge representation:** To store what it knows and hears.
- **Automated reasoning:** Answer the Questions based on the stored information.
- **Machine learning:** Must be able to adapt in new circumstances.

Acting Humanly: The Turing test approach

Total Turing Test:

In order to pass total turing test, it requires interaction with objects and people in the real world. A robot will need

- computer vision and speech recognition to perceive the world;
- robotics to manipulate objects and move.

Reverse Turing Test

flips the roles. Instead of a human judging a machine, a machine judges a human (e.g., CAPTCHA)

The primary reason is Security via "Proof of Personhood."

Acting Humanly: The Turing test approach

Loebner Prize:

- Every year in Boston
- In 2019, the chatbot were judge by the public

"What if people call a human a machine?", "Shakespeare expert," "Machine did not win... the human failed"

This shows famous failure of the test's methodology.

Thinking humanly: The cognitive modeling approach

- ❖ **Cognitive:** It is the mental process act in brain for acquiring knowledge, and analyse the information. For example: Thinking, knowing, Remembering etc..
- ❖ In order to make program think like human we should know how exactly our brain processes:
 - introspection—trying to catch our own thoughts as they go by;
 - psychological experiments—observing a person in action
 - Brain imaging—observing the brain in action.
- ❖ **Cognitive Science:**
 - Hard to understand how human thinks
 - justifications or reasons for a decision, irrational human behavior
- ❖ Human have hardware of brain, the machine do not have the same hardware as humans. So it is hard to claim that we can build machine that can think like human

Thinking rationally: The “laws of thought” approach

- ❖ The Greek philosopher Aristotle was one of the first who attempt to codify the **right thinking** that is irrefutable reasoning process.

Syllogism:

- Premise 1: All men are mortal.
 - Premise 2: Aryan is a man.
 - Conclusion: Therefore, Aryan is mortal.
- ❖ Problems:
 1. The Translation Problem (Informal vs. Formal)
 - It is incredibly difficult to translate real-world messiness (informal knowledge) into strict code (formal logic).
 - Logic requires 100% certainty (True/False), but the real world is full of uncertainty (e.g., "maybe," "probably," "sometimes").

Thinking rationally: The “laws of thought” approach

- ❖ Problems:

- 2. The Computational Problem (Theory vs. Practice)

- ❖ Knowing how to solve a problem in theory is different from actually solving it.
 - ❖ Even a problem with just a few dozen facts can create trillions of logical possibilities. Without specific guidance, a computer will run out of memory or take forever trying to check them all.

Acting Rationally: The Rational Agent Approach

- ❖ This is the most appropriate approach to AI.
- ❖ **Rational Behavior:** Acting rationally means acting to achieve the best outcome (or the best expected outcome if there is uncertainty).
- ❖ **The right thing:** that which is expected to maximize goal achievement, given the available information.
- ❖ **Rational Agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

Advantages:

1. It is Scientifically Measurable
2. It Handles Uncertainty

Problem: Computational Limit, Pure Rationality,

Foundation of AI

Philosophy: Logic, reasoning, mind as a physical system, foundations of learning, language and rationality.

- Where does knowledge come from?
- How does knowledge lead to action?
- How does mental mind arise from physical brain?
- Can formal rules be used to draw valid conclusions?

Foundation of AI

Mathematics: Formal representation and proof algorithms, computation, undecidability, intractability, probability.

- What are the formal rules to draw the valid conclusions?
- What can be computed?
- How do we reason with uncertain information?

Foundation of AI

Economics: Formal theory of rational decisions, game theory, operation research.

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in future?

Foundation of AI

Psychology: Adaptation, phenomena of perception and motor control.

- How humans and animals think and act?

Linguistics: Knowledge representation, grammar

- How does language relate to thought?

Neuroscience: Physical substrate for mental activities

- How do brains process information?

Computer Engineering:

- How do we build efficient computers?

AI Tree

Artificial Intelligence (AI) is a broad field that encompasses various subfields and branches, each focusing on different aspects of creating intelligent machines. Here are the main branches of AI:

Main branches of AI



AI Branches

❖ Machine Learning

- Machine Learning is the study of computer algorithms that improve automatically through experience and data.
- **Applications:** Image and speech recognition, predictive analytics, recommendation systems.

❖ Natural Language Processing

- Definition: NLP focuses on the interaction between computers and human languages, enabling machines to understand, interpret, and generate human language.
- **Applications:** Language translation, sentiment analysis, chatbots, text summarization.

❖ Computer Vision (CV)

- Definition: Computer Vision involves enabling machines to interpret and make decisions based on visual data.
- **Applications:** Facial recognition, autonomous vehicles, medical image analysis, surveillance.

AI Branches

❖ Robotics

- Definition: Robotics is the branch of AI concerned with the design, construction, operation, and use of robots.
- **Applications:** Manufacturing automation, medical robots, exploration (space, underwater), service robots.

❖ Expert Systems

- Definition: Expert Systems are AI programs that simulate the decision-making ability of a human expert.
- **Applications:** Medical diagnosis, financial forecasting, troubleshooting systems.

❖ Deep Learning

- Definition: Neural Networks are computing systems inspired by the biological neural networks of animal brains, and Deep Learning is a subset of ML that uses multi-layered neural networks.
- Applications: Image and speech recognition, natural language processing, game playing.

AI Branches

❖ Fuzzy Logic

- Definition: Fuzzy Logic is a form of many-valued logic that deals with reasoning that is approximate rather than fixed and exact.
- **Applications:** Control systems, decision-making systems, pattern recognition.

❖ Evolutionary Computation

- Definition: Evolutionary Computation is a family of algorithms for global optimization inspired by biological evolution.
- **Applications:** Optimization problems, artificial life, evolutionary robotics.

❖ Swarm Intelligence

- Definition: Swarm Intelligence involves the collective behavior of decentralized, self-organized systems, typically made up of a population of simple agents.
- **Applications:** Optimization problems, robotics, network routing.

❖ Cognitive Computing

- Definition: Cognitive Computing refers to systems that mimic human thought processes in a computerized model.
- **Applications:** Healthcare diagnostics, financial analysis, customer service.

AI–Applications in Interdisciplinary Areas

1. Health Care

- Medical Imaging: AI can analyze medical images (like X-rays, MRIs, and CT scans) to assist in diagnosing conditions such as cancer, heart disease, and neurological disorders.
- Predictive Analytics: AI helps in predicting patient outcomes, readmission rates, and identifying high-risk patients.
- Drug Discovery: Machine learning algorithms are used to analyze chemical compounds and predict their efficacy, helping to accelerate the drug discovery process.
- Personalized Medicine: AI is used to tailor treatments based on an individual's genetic profile and medical history.

AI–Applications in Interdisciplinary Areas

2. Finance

- Fraud Detection: AI systems monitor transactions in real-time to detect unusual patterns that may indicate fraudulent activity.
- Algorithmic Trading: AI is used in financial markets to analyze large datasets and make real-time trading decisions.
- Customer Service: AI-driven chatbots and virtual assistants are used for handling customer inquiries and managing financial services.

AI–Applications in Interdisciplinary Areas

3. Autonomous Vehicles

- **Self-Driving Cars:** AI is used for the perception, decision-making, and control systems in autonomous vehicles. It helps vehicles navigate traffic, avoid obstacles, and make real-time decisions based on sensor data.
- **Traffic Management:** AI can optimize traffic flow by analyzing patterns and adjusting signals in real-time.

AI–Applications in Interdisciplinary Areas

4. Retail and E-commerce

- **Personalized Recommendations:** AI is used by platforms like Amazon and Netflix to analyze user behavior and recommend products, movies, or music.
- **Collaborative filtering** focuses on user similarity to recommend items, content-based filtering recommends items exclusively according to item profile features.
- **Inventory Management:** AI helps retailers predict demand, optimize stock levels, and streamline supply chains.
- **Customer Support:** AI-powered chatbots assist customers in finding products, answering questions, or resolving issues without human intervention.

AI–Applications in Interdisciplinary Areas

5. Manufacturing

- Predictive Maintenance: AI can monitor machinery and equipment to predict when they are likely to fail, reducing downtime and improving efficiency.
- Quality Control: AI systems can detect defects in products during production using image recognition and other techniques.
- Supply Chain Optimization: AI helps optimize logistics and production schedules, reducing waste and improving efficiency.

AI–Applications in Interdisciplinary Areas

6. Natural Language Processing (NLP)

- Language Translation: AI systems like Google Translate help translate text and speech in real-time across multiple languages.
- Sentiment Analysis: AI can analyze text data (such as customer reviews or social media posts) to determine the sentiment behind it.
- Speech Recognition: AI is used in voice assistants (like Siri, Alexa, and Google Assistant) to understand and process spoken commands.

AI–Applications in Interdisciplinary Areas

7. Education

- Personalized Learning: AI-powered platforms can tailor educational content to an individual's learning style, pace, and preferences.
- Automated Grading: AI systems are used to grade assignments, essays, and tests, providing instant feedback to students.
- Tutoring and Assistance: Virtual tutors powered by AI can help students with specific problems, answering questions and providing explanations in real-time.

AI–Applications in Interdisciplinary Areas

8. Entertainment and Media

- Content Creation: AI can generate art, music, and even entire articles based on prompts or historical data, assisting in creative processes.
- Video Editing: AI tools can automate tasks like color correction, scene transitions, and even generate video content from raw footage.

AI–Applications in Interdisciplinary Areas

9. Law and Legal Services

- Contract Review: AI tools can quickly analyze legal documents and contracts, highlighting key clauses, risks, and inconsistencies.
- Predictive Analytics: AI is used to predict the outcomes of legal cases based on historical data and case precedents.

AI–Applications in Interdisciplinary Areas

Others:

- Agriculture
- Marketing and Advertising
- Governance
- Energy
- Human resources
- Others

Learning

- Learning is the active process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

While "knowledge" is the destination, learning is the journey. It is the bridge between "not knowing" and "knowing."

- What does it involve?
 - **Attention:** You cannot learn what you do not focus on.
 - **Encoding:** takes the new information and translates it into a format it can understand
 - **Association:** New information is rarely learned in isolation. It involves connecting new data to existing mental frameworks
 - **Feedback:** Experience the result of an action (success or failure) and adjust.

Knowledge

- ❖ Knowledge is a theoretical or practical understanding of a subject or a domain.
- ❖ Knowledge is also the sum of what is currently known.
- ❖ Knowledge is “the sum of what is known: the body of truth, information, and principles acquired by mankind.” Or, “Knowledge is what I know, Information is what we know.”

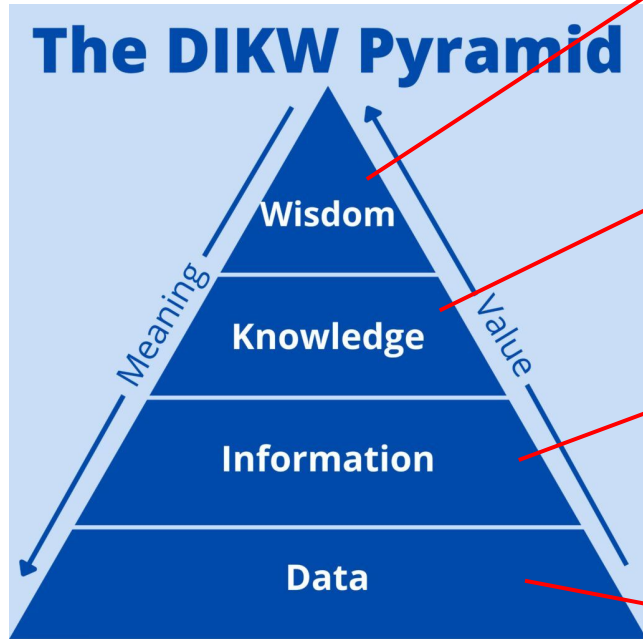
Formal Definition

- ❖ Knowledge is “information combined with experience, context, interpretation, and reflection. It is a high-value form of information that is ready to apply to decisions and actions.” (T. Davenport et al., 1998).
- ❖ Knowledge is “human expertise stored in a person’s mind, gained through experience, and interaction with the person’s environment.” (Sunasee and Sewery, 2002)

Importance of Knowledge

- **Speed and Efficiency:** Knowledge allows you to skip the "trial and error" phase. You don't have to test which berries are poisonous; you use the knowledge passed down to you to survive.
- **Problem Solving:** You cannot solve a problem if you don't understand the variables. Knowledge provides the context needed to diagnose issues.
- **Innovation:** New ideas are rarely 100% original. They are usually a unique combination of two existing pieces of knowledge.

DIKW Pyramid



Wisdom is the highest level of the DIKW pyramid, representing the ability to apply knowledge and experience to make sound judgments and decisions.

Knowledge is the understanding gained from information, through analysis, interpretation, and synthesis.

Data becomes information when it is organized, processed, and interpreted in a meaningful way.

Data is a collection of facts in a raw or unorganized form such as numbers or characters.

State of the Art

"State of the Art" refers to the highest level of development reached at a specific time.

"State of the Art" by field

1. Robotics & Autonomous Vehicles
 - a. Self-Driving Cars:
 - i. **Previous:** Testing Phase. Waymo had driven 10 million miles. Human safety drivers were constantly needed
 - ii. 2025: Tesla's FSD (Full Self-Driving) is widely deployed with supervised autonomy.
 - b. Robots (Legged):
 - i. **Previous:** Robots like BigDog and Atlas could do backflips but were loud, hydraulic, and strictly research projects.
 - ii. 2025: Tesla Optimus (Gen 3) and Figure 02.

2. Medicine & Science

a. Protein Folding

- i. **Previous:** Unsolved. Predicting how proteins fold was a 50-year-old "grand challenge" in biology.
- ii. **2025:** Solved (Nobel Prize). AlphaFold 3 (Google DeepMind) predicts the structure of virtually all of life's molecules (DNA, RNA, Ligands).

b. Diagnostics

- i. **Previous:** AI equaled doctors in analyzing static images (X-rays, skin cancer).
- ii. **2025:** AI now outperforms specialists in complex diagnostics (e.g., metastatic cancer detection).

3. Language & Creativity (Generative AI)

- a. **Text: Previous:** GPT-2 could write a few coherent paragraphs but lost the plot quickly.

2025: Reasoning Agents, GPT3, Gemini 3.0 and others

State of the Art

Video:

- i. **Previous:** AI video was non-existent or limited to blurry, silent, 2-second clips.
- ii. **2025:** Cinematic Reality. Models like Sora 2 and Veo 3 generate 1080p, 60-second movie clips with synchronized audio

4. Environment & Climate

a. Weather:

- i. **Previous:** Traditional. Relied on massive supercomputers solving physics equations
- ii. **2025:** AI-Native. AI models like GraphCast predict extreme weather (hurricanes, heatwaves) faster and more accurately

State of the Art

5. Coding & Software Engineering

a. Coding:

- i. **Previous:** Autocomplete. Tools could suggest the next line of code
- ii. **2025:** Autonomous Agents. AI Software Engineers (e.g., Devin, SWE-agent) can take a GitHub issue, browse the codebase, write a fix, and run tests autonomously.

What Humans and AI can do?

- AI is a Specialist Engine: It excels at processing massive data, executing specific rules, and scaling outputs (speed/volume).
- Human is a Generalist Architect: We excel at navigating ambiguity, making ethical judgments, understanding context, and managing the AI itself.

Comparison by Parameter for Human and AI

Parameter	Humans	AI	
Speed & Scale	Slow & Linear. We can read limited amount of content and write one draft at a time.	Instant & Infinite. Can process 10 million words	AI
Reasoning (Math/Science)	Deep & Novel. Humans invent new math theories. However, calculation speed is limited.	Gold-Medal Level. Models like OpenAI o1 and Gemini 2.0 now score 90%+ on PhD-level science (GPQA) and win Gold Medals in Math Olympiads.	Both
Creativity	Intentional & Emotional. Art comes from lived experience	Generative. Can produce technically perfect 1080p video (Sora) or poetry, but it is "remixing" existing patterns.	Human

Comparison by Parameter for Human and AI

Parameter	Humans	AI	
Physical World	Highly Adaptable. You can walk into a messy room, step over a toy, and fold laundry without thinking	Improving but Rigid. Robots like Tesla Optimus can now fold laundry, but they are slower and struggle if the environment changes unexpectedly.	Human
Reliability	Variable. Humans get tired, bored, or distracted. Performance fluctuates.	Consistent. AI never sleeps, never gets bored, and follows instructions perfectly (once trained).	AI
Adaptability	Universal. A human can learn to play chess, cook, and drive in one lifetime.	Brittle. An AI trained for chess cannot cook. Even "General" models struggle when rules change suddenly (e.g., the ARC-AGI benchmark).	Human

Agent

Agent:

- An **agent** is anything that can be viewed as perceiving its environment through **sensors** and **acting** upon that environment through **actuators**.
1. **Environment:** The area around the agent that it interacts with.
 2. **Sensors:** Sensors are tools that AI agent uses to perceive their environment.
 3. **Actuators:** Tools uses to interact with their environment through some actions.
 4. **Effectors:** Action perform through decision made

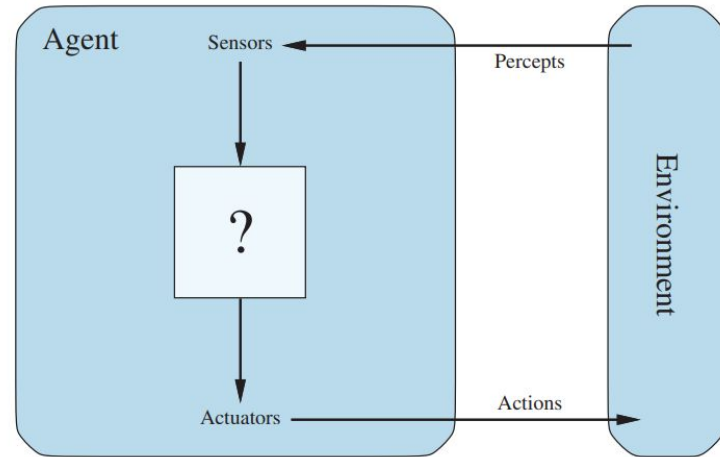


Figure. Agents interact with environments through sensors and actuators.

Intelligent Agents

- ❖ An **Intelligent Agent** is a broad term for any system that can perceive its environment and act upon it.
- ❖ **Example:** A robotic vacuum is an intelligent agent. It senses dirt and moves to clean it.
- ❖ **Importance:**
 - Handling Complexity and Data
 - Continuous Learning (Adaptability)
 - Automation of Decision Making
- ❖ **Rational Agent:** A rational agent is one that always selects the action that is expected to maximize its Performance Measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent possesses.

https://www.youtube.com/shorts/D_EYMga_jJQ

Intelligent Agents

❖ Applications of Intelligence Agent

- Autonomous Systems
- Personal Assistants
- Recommendation Systems
- Financial Trading

❖ Challenges for Intelligent Agents

- Ethical and Legal Implications
- Robustness and Reliability
- Interpretability
- Scalability and Efficiency

PEAS Framework

Things that should be consider building Agent.

- **Performance Measure:** How do we define success? (e.g., Safety, Speed, Cleanliness).
- **Environment:** Where does it operate? (e.g., Highway, Kitchen).
- **Actuators:** What can it use to act? (e.g., Steering wheel, Robotic arm).
- **Sensors:** How does it perceive? (e.g., Camera, GPS).

Factors to Select the Agent

❖ Environmental Factors

- Can the agent see the entire world at once (Fully) or only a part of it (Partially)?
- Is the outcome of an action guaranteed (Deterministic) or random (Stochastic)?
- Does the environment change while the agent is thinking (Dynamic) or stay still (Static)?

❖ Task Complexity

- Does the agent need to plan long-term steps (like Chess), or is every action isolated (like identifying a photo)?
- Is the goal a simple "Pass/Fail," or does it require balancing trade-offs (e.g., speed vs. safety)?
- Does the agent work alone, or must it compete/cooperate with other agents and humans?

Factors to Select the Agent

❖ Resource Constraints

- Must the agent react in milliseconds (Reflex), or can it take time to think (Complex Planning)?
- Does the hardware allow the agent to remember a long history, or is memory limited?
- Is the agent battery-operated (requires efficiency) or connected to unlimited power?

❖ Adaptability

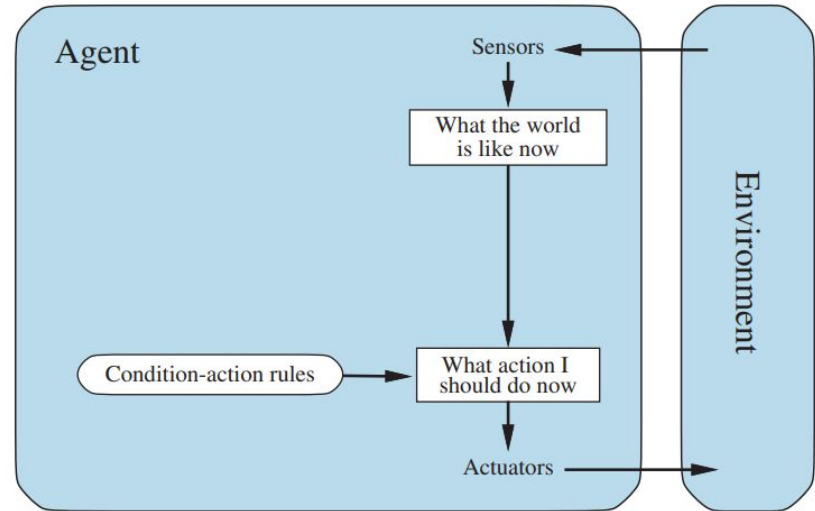
- Does the agent need to improve its own performance over time without human updates?
- Can the agent keep working if a sensor breaks or sends bad data?
- Can the agent handle a brand-new situation it has never seen before?

Type of Agents

❖ Simple Reflex Agents

- The simplest kind of agent is the simple reflex agent.
- These agents select actions on the basis Simple reflex agent of the current percept, ignoring the rest of the percept history.
- Speed, Simplicity, Efficiency
- If the agent needs to remember something from 5 seconds ago to make a decision, this agent will fail.

- ❖ **Example:** Traffic light control systems that change signals based on fixed timing.

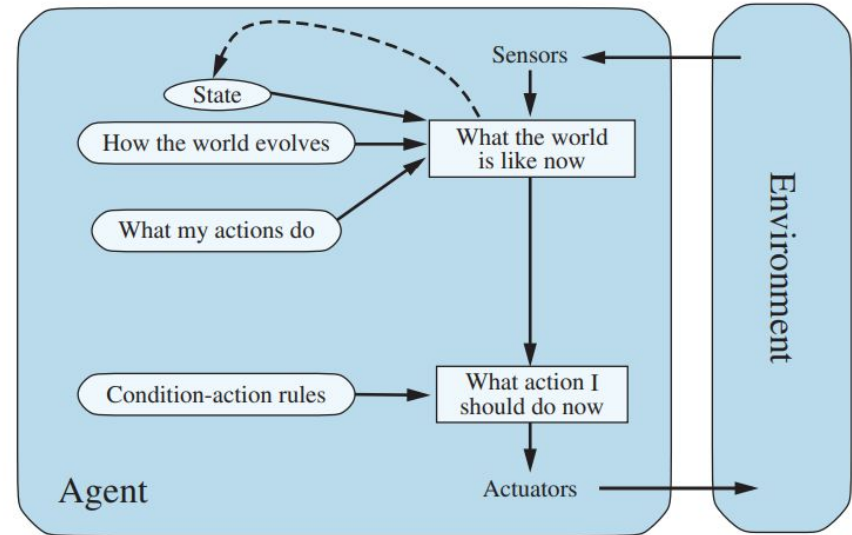


Type of Agents

❖ Model-Based Reflex Agents

- The Agent with Memory" These agents have an internal state (memory) to keep track of the world.
- Deal with partial observability and dynamic changes more effectively
- Maintaining an internal model increases computational demands which requires more memory and processing power to track changes in the environment.
- They are beneficial in situations where the environment is dynamic and not all elements can be directly observed at once.

- ❖ **Example:** Robot vacuum cleaners that map rooms and tracks cleaned areas.

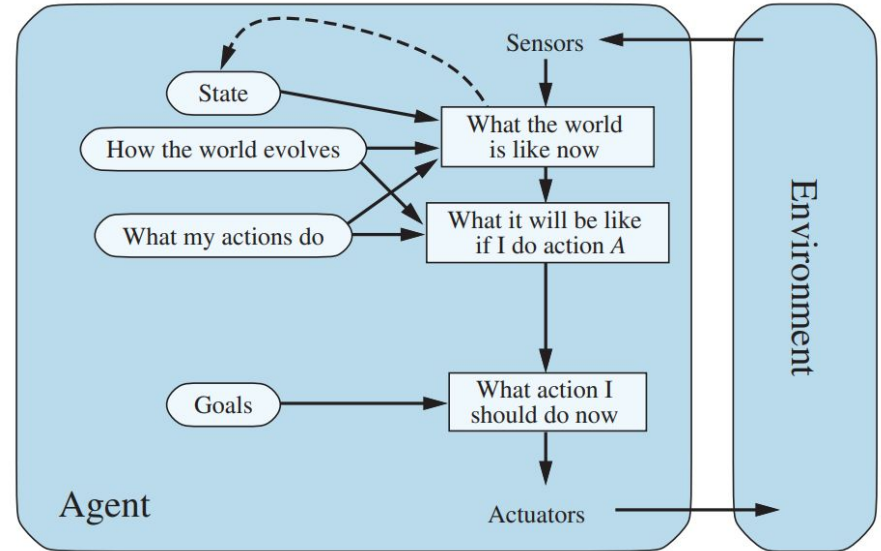


Type of Agents

❖ Goal-Based Agents

- Goal-based agents select actions by considering future states relative to explicit goals.
- They often use planning algorithms that explore multiple possible actions to find the most effective sequence of steps that lead to their goal.
- If conditions change or new information arises, it can re-plan and adjust their strategies
- It predict future outcomes to find the best course of action.

- ❖ **Example:** Logistics routing agents that find optimal delivery routes based on factors

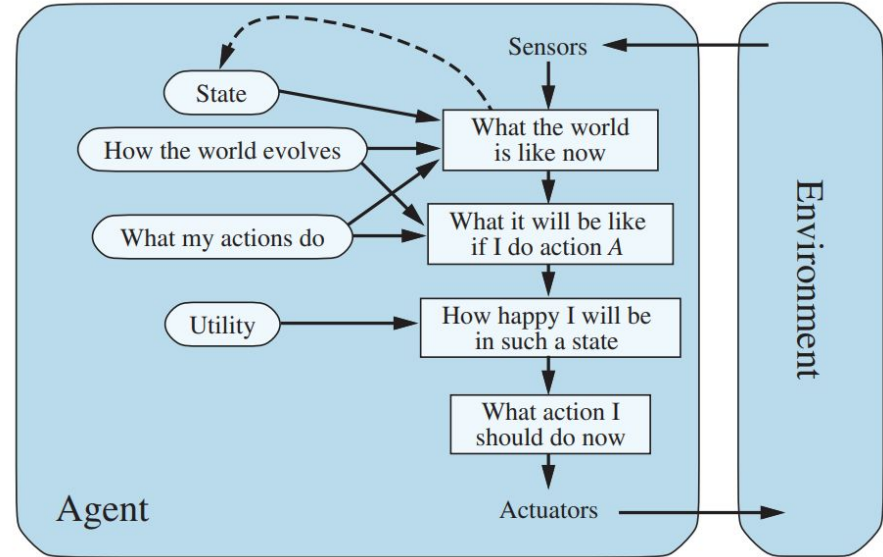


Type of Agents

❖ Utility-Based Agents

- This agent doesn't just want to reach the goal
- It wants to reach it in the best way possible
- A utility function to quantify preferences and make trade-offs between competing objectives, enabling nuanced decision-making in uncertain or resource-limited situations.
- Finding utility functions for different factors can be computationally intensive and complex.

- ❖ **Example:** Financial portfolio management agents that evaluate investments

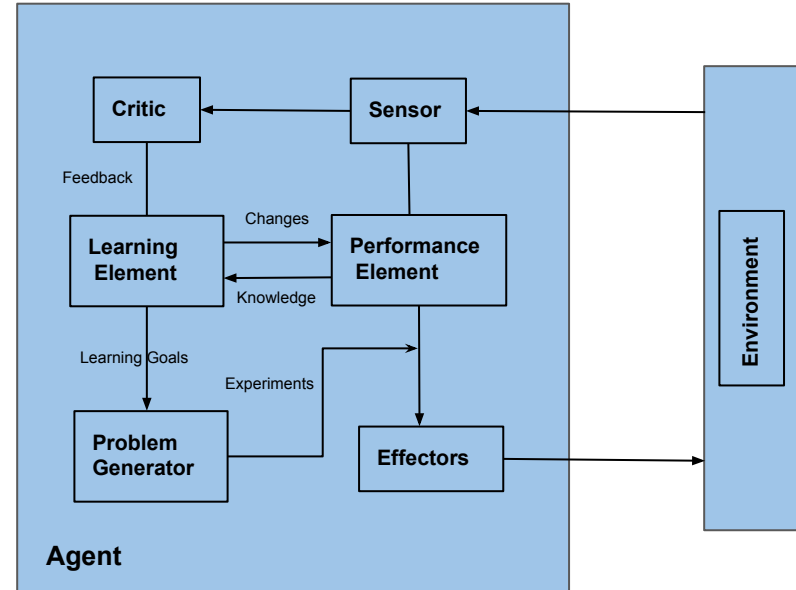


Type of Agents

❖ Learning Agents

- It improve their performance over time by learning from experience and updating their internal models, strategies or policies.
- Adapt to changes in the environment and often outperform static agents in dynamic contexts.
- It can apply lessons learned in one context to new, similar situations enhancing their versatility.

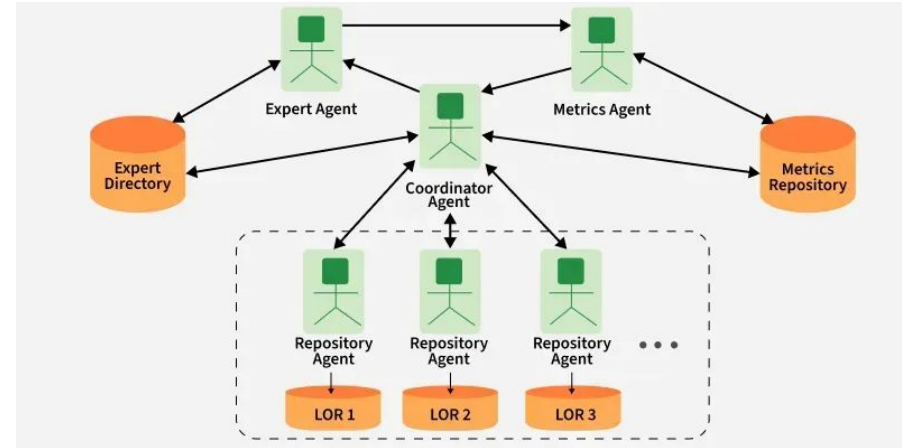
- ❖ **Example:** Customer service chatbots can improve response accuracy over time by learning



Type of Agents

❖ Multi-Agent Systems

- Multiple agents share the same environment, either cooperating or competing to achieve individual or group goals.
- Decentralized, often requiring communication, negotiation or coordination protocols.
- They are ideal for decentralized environments like traffic control, robotics or large-scale simulations where agents need to collaborate or make decisions independently.

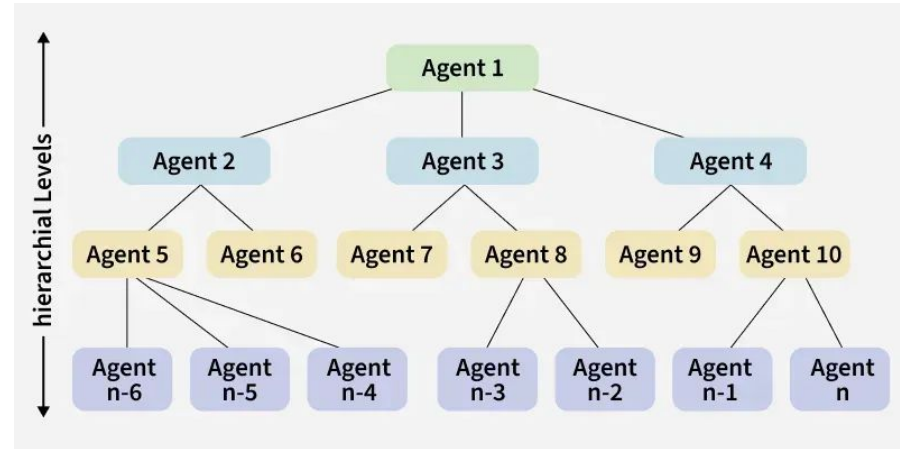


Type of Agents

❖ Hierarchical Agents

- Hierarchical agents organize behavior into multiple layers such as strategic, tactical and operational.
- Higher levels make abstract decisions that break down into more specific subgoals for lower levels to execute.
- Complex tasks are broken down into simpler subtasks.

- ❖ **Example:** Drone delivery systems in which fleet management is done at top level and individual navigation at lower level.



Agentic AI

Agentic AI refers to AI systems designed to show "agency"—the ability to actively pursue complex goals, break them down into sub-tasks, and execute them using external tools.

❖ How is this different from **Previous AI**

Features	Previous AI	Agentic AI
Primary Function	Generate text/images	Execute tasks/workflows
Interaction	Conversation (Chat)	Command for the Improvement
Environment	It is connected to the web, but mostly in Read-Only mode.	It has Read-Write access. It can change the state of the world.
If it fails	It hallucinates or apologizes	It self-corrects and tries a new way

Characteristics of Agentic AI

❖ Proactive

- It doesn't wait to be told. It looks for problems before they happen.
- It uses historical data and trends to anticipate future needs or bottlenecks
- It identifies and solves potential issues immediately to prevent them from escalating into critical failures.

❖ Adaptable

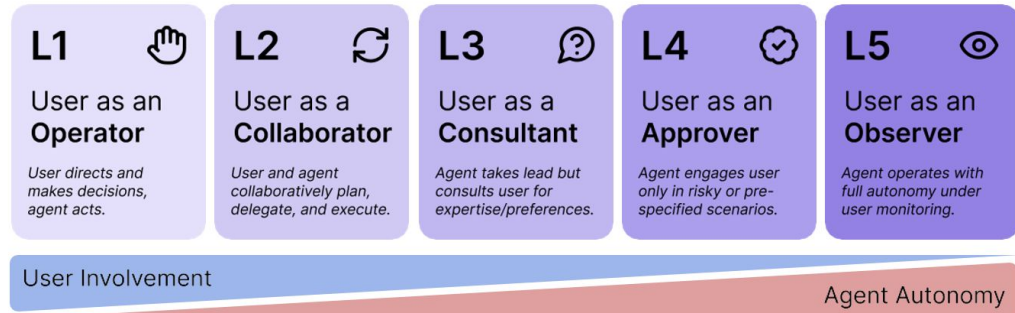
- It understands the specific rules, tone, and regulations of its domain
- It changes its strategy in real-time based on new inputs; it does not follow a rigid, linear script.
- It can navigate "edge cases" or unique scenarios that standard, pre-programmed software would simply reject as errors.

Characteristics of Agentic AI

- ❖ Collaborative
 - It is designed to function as a "co-pilot," capable of asking for human approval on high-stakes decisions rather than working in a black box.
 - It acts as a node in a network, sharing data and status updates with other independent agents to ensure everyone is on the same page.
 - It interprets the intent of other team members (human or AI) to ensure its actions support the broader, shared objective.
- ❖ Specialized
 - Instead of being "okay" at everything, it focuses on being "perfect" at one specific task
 - It breaks a massive, complex goal into smaller sub-tasks and routes them to the specific agents best suited to handle them.
 - It operates like a department of specialists (a coder, a writer, a lawyer) rather than one overworked general manager.

Autonomous Intelligence

- ❖ Autonomous Intelligence represents the highest level of AI maturity.
- ❖ It refers to systems that can operate independently in open-ended environments to achieve high-level objectives without constant human guidance or intervention.
- ❖ Level of Autonomous
- ❖ Examples:
 - NASA's Perseverance Rover
 - Amazon "Proteus" Warehouse Robots
 - Waymo Robotaxis



THE END!!